Jiani (Elva) He

Data Science Consultant · Independent Researcher
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RESEARCH INTERESTS

AI-assisted reverse logistics and resource recovery; inspection and disposition under uncertainty; causal decision support; forecast stability and planning value; AI deployment assurance.

Related areas on the U.S. Critical and Emerging Technologies List (2026): AI and Autonomy (research), and Semiconductors and Microelectronics (through data center hardware supply chains).

PROFESSIONAL EXPERIENCE

Accenture Jun 2024 - Present
Data Science Consultant Seattle, Washington
Forward and reverse logistics in a technology client's data center supply chain. Promoted from Senior Analyst in 2024.

- Lead planning analyses for the client and present recommendations directly to senior leaders.
- Steer recovery decisions for retired hardware: analyze reverse-logistics cycle time and on-time delivery in SQL, find bottlenecks, and route assets to reuse, resale, recycling, or disposal by value and lifecycle stage.
- Rebalance inventory across sites with an optimization engine, matching stock targets to demand to prevent shortages.
- Plan quality-assurance component quantities from demand and bill-of-materials data; define service metrics and support automated inventory adjustments.
- Bring generative AI into daily analysis: standardize prompts and queries for enterprise LLM assistants, write reusable guides, and train analysts.
- Mentored five junior analysts across both roles; lead onboarding and team training.

Accenture Oct 2021 - May 2024
Data Science Senior Analyst Seattle, Washington

- Built SQL dashboards and shortage alerts that compare inventory with demand and inform transfers between sites.
- Tracked reverse-logistics cycle times and on-time delivery; flagged bottlenecks and supported reuse, disposal, and recovery scheduling.
- Diagnosed inventory imbalances and abnormal cycle times with statistical analysis; automated reporting and purchase-order workflows.

BMW Group Feb - Aug 2021
Data Scientist Intern · MIT Capstone, Dynamic Vehicle Pricing Cambridge, Massachusetts

- Modeled vehicle demand and cross-price elasticity in Python to recommend discounts by customer segment and sales channel; built Tableau pricing dashboards.

EDUCATION

Massachusetts Institute of Technology Aug 2021
MS, Business Analytics · Operations Research Center | GPA 4.7 / 5.0 Cambridge, Massachusetts
Classification trees for 40,000 news articles (Python/Julia) and robust bike-allocation models (Julia). Analytics Lab: an NLP admissions Q&A pipeline for AdmitHub with response evaluation and human review.

Purdue University May 2020

BS, Industrial Engineering · Honors & Distinction | GPA 3.9 / 4.0 West Lafayette, Indiana

LSTM forecasting models (Keras) and reinforcement learning for sensor placement (Python/R).

RESEARCH & PUBLICATIONS

The conferences listed below have concluded; proceedings publication is pending.

Lead-author papers

1. **Jiani He**, Dingyan Shang, Yihua Xu, Shiqi Huang, Yan Lyu, Jize Li, Shangjing Tang. 2026. "Semantic Signal-Assisted Inspection and Recovery Allocation in Reverse Logistics." *4th International Conference on Artificial Intelligence, Blockchain, and Internet of Things (AIBThings 2026)*. arXiv:2609.02116. **AIBThings 2026; proceedings forthcoming**

2. Shiqi Huang*, **Jiani He***, Dingyan Shang*, Yihua Xu, Jize Li, Yan Lyu, Lashimi Muraleedharan Nair (*equal contribution). 2026. "DACRI: Decision-Aware Causal Intervention Ranking for Critical Supply Chains." *International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME 2026) · 15–17 October 2026 · Bali, Indonesia*. arXiv:2608.11154. **ICECCME 2026; proceedings forthcoming**

Conference papers

3. Yihua Xu, Yan Lyu, Jize Li, **Jiani He**, You Lin, Yidan Hu, Rubing Qiang. 2026. "When Does an Added Planning Capability Create Operational Value? An Observable Evidence Standard." *3rd World Congress on Industrial Engineering and Operations Management · Irvine, California*. arXiv:2610.27305. **IEOM Irvine 2026; proceedings forthcoming**

4. Yan Lyu, Yihua Xu, **Jiani He**, Jize Li, Rubing Qiang, Yidan Hu. 2026. "From Weak Signals to Supply Chain Reliability Risk: A Feedback-Driven Risk Accumulation Framework for Late-Delivery Prediction, Early Warning, and Risk Prioritization." *3rd World Congress on Industrial Engineering and Operations Management · Irvine, California*. SSRN preprint submitted · posting pending. **IEOM Irvine 2026; proceedings forthcoming**

5. Jize Li, **Jiani He**, Dishu Yang, Dingyan Shang, Jingjing Liu, Shiqi Huang. 2026. "Accuracy-Preserving Stability Regularization for Large-Scale Retail Demand Forecasting." *ICEME 2026*. arXiv:2607.13331. **ICEME 2026; proceedings forthcoming**

6. Yihua Xu, **Jiani He**, Ishita Chirag Talati, Yitian Qian, Youting Wang, Zhenyu Xu, Dingyan Shang. 2026. "Policy-Bounded, Provenance-Aware Deployment Assurance for LLM Request Routing." *ICOAI 2026 · public preprint on SSRN*. SSRN 7554440. **ICOAI 2026; proceedings forthcoming**

SERVICE & MENTORING

- Peer reviewer for 10+ journal manuscripts.
- Mentored five junior analysts across Accenture roles; led onboarding and team training, including AI-assisted analysis.

SKILLS

Tools: SQL, Python, R, Julia, Keras, Tableau, AWS

Modeling: Machine learning, Forecasting, Regression, Classification, Clustering, Robust optimization

Analytics & BI: Statistical analysis, Root-cause analysis, Data visualization, Dashboards, KPI design

AI: Generative AI, NLP, LLM workflows, Prompt & query design

Consulting: Problem framing, Client communication, Stakeholder coordination, Project delivery, Mentoring